random variables. Assuming that the correlation is positive (let's not worry about negative c):

- If this correlation is high, then the two random variables X_i and X_j are said to be correlated. This means if the value of one increases, the value of the other also tends to increase—if the value of one decreases, the value of the other will decrease. Our neighbor Alice's houses and our own are an example of correlated variables.
- On the other hand, if this correlation is low and close to zero, then whatever the value of X_i is, what we know about the value of X_j does not change. This is because there is no correlation between the two variables. Alix's house in California and our house fall into this category.

To make it more concrete, let's define a multivariate Gaussian distribution that jointly models three random variables, the price of our house A, the price of our neighbor Alice's house B, and the price of Alix's house in California C.

NOTE **It's usually convenient to assume that the mean vector of this Gaussian is normalized to be the zero vector. This is in fact what's usually done in practice, as it simplifies a lot of mathematical details.**

This three-dimensional Gaussian also has a covariance matrix described in Figure 2.4.

Once again, the diagonal cells tell us the variances of individual random variables. B has a slightly larger variance (3) than A (1), meaning that we are more uncertainty about the value of B, due to the fact that we don't know everything about our neighbor's house, so a more accurate estimate cannot be done. The third variable C, on the other hand, has the largest variance to denote the fact that houses in California have an overall wider price range.

SIDEBAR **A large variance indicates more uncertainty**

The values that are being used here (1, 3, 10) are example values to make the point that the larger a variance of a random variable is, the more uncertain we are about the value of the variable (before learning what its value is).

Further, the covariance between our house A and our neighbor's B is 0.9, which indicates that the prices of the two houses are correlated in a non-negligible way. This makes sense, since if we know the price of our neighbor's house, we will be able to obtain a better estimate for our own house that is on the same street. Also notice that neither A nor B has any correlation with the price of the house in California, since location-wise, C has nothing in common with A or B. Another way to say this is even if we know how much the house in California costs, we won't learn anything about the price of our own house.

Let's now visualize this three-dimensional Gaussian using a parallel coordinates plot in Figure 2.5.

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